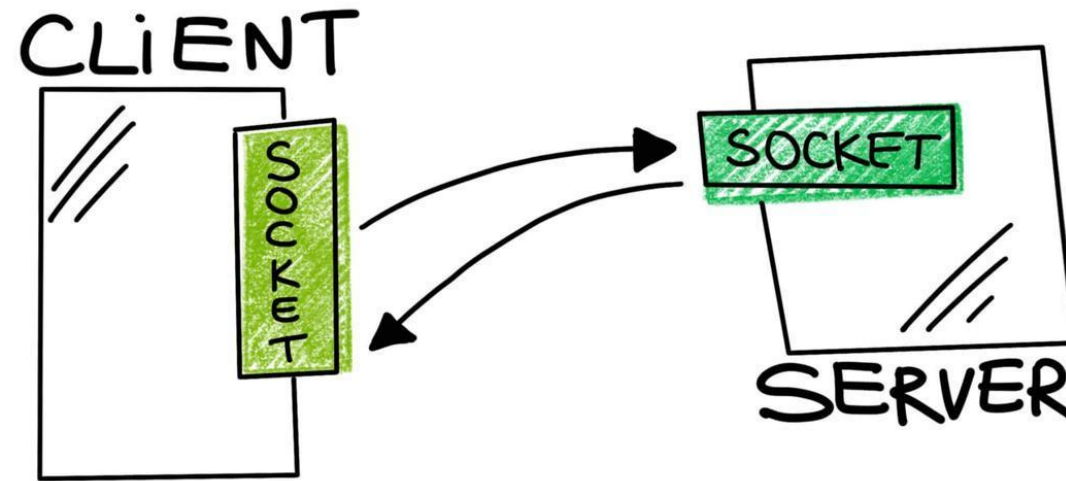




Socket Programming Basics

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Sockets



- A socket is an **endpoint** to receive and send information over the network (defined by a **file descriptor**)
- Sockets allow communication between different processes.

Types of sockets

- **Stream socket** :(connection-based socket)
 - Reliable, error free, no out of order packets (uses TCP)
 - Applications: ssh, http, text communication (WhatsApp, Instagram,...), FTP , Simple Mail Transfer Protocol (SMTP) (such as Gmail, Outlook, ...)
- **Datagram socket** (connectionless socket)
 - It provides unreliable, best-effort networking service
 - Packets may be lost; may arrive out of order (uses UDP)
 - It is faster and simpler
 - Applications: online games (since UDP is faster and games don't tolerate delay) , Video Conferencing (skype, zoom, ...) , Voice Over IP (VoIP) (WhatsApp calls, ...)

File Descriptors

File descriptor values for the standard I/O streams

File descriptor value	Standard stream	Mode	Endpoint
0	stdin	read	terminal
1	stdout	write	terminal
2	stderr	write	terminal



Sockets

A socket is an end point determined by two things:

- IP address
- Port number

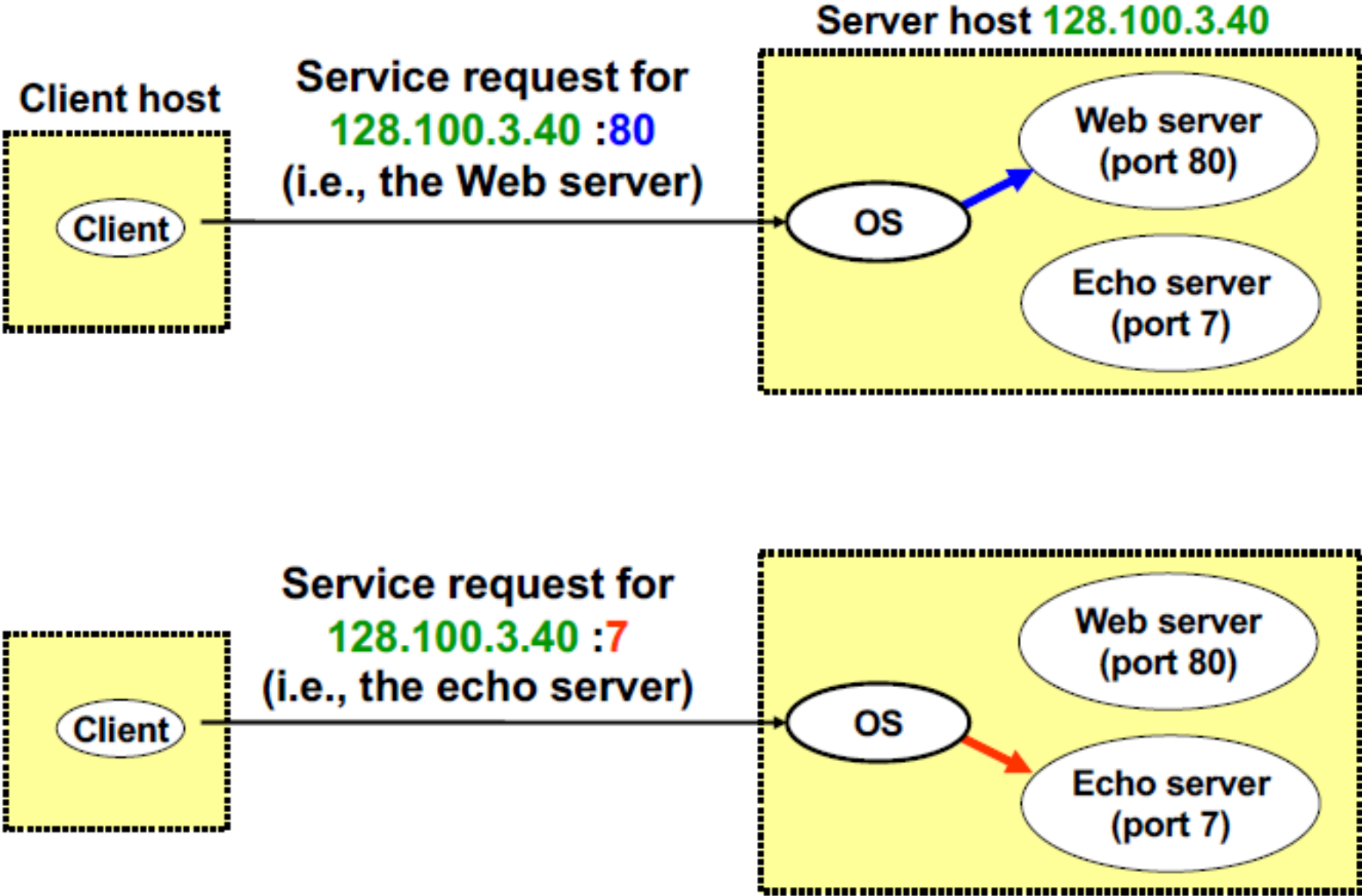
Think of your machine as an apartment building:

- A **port** is an apartment number.
- A **socket** is the door of an apartment.
- An **IP address** is the street address of the building.

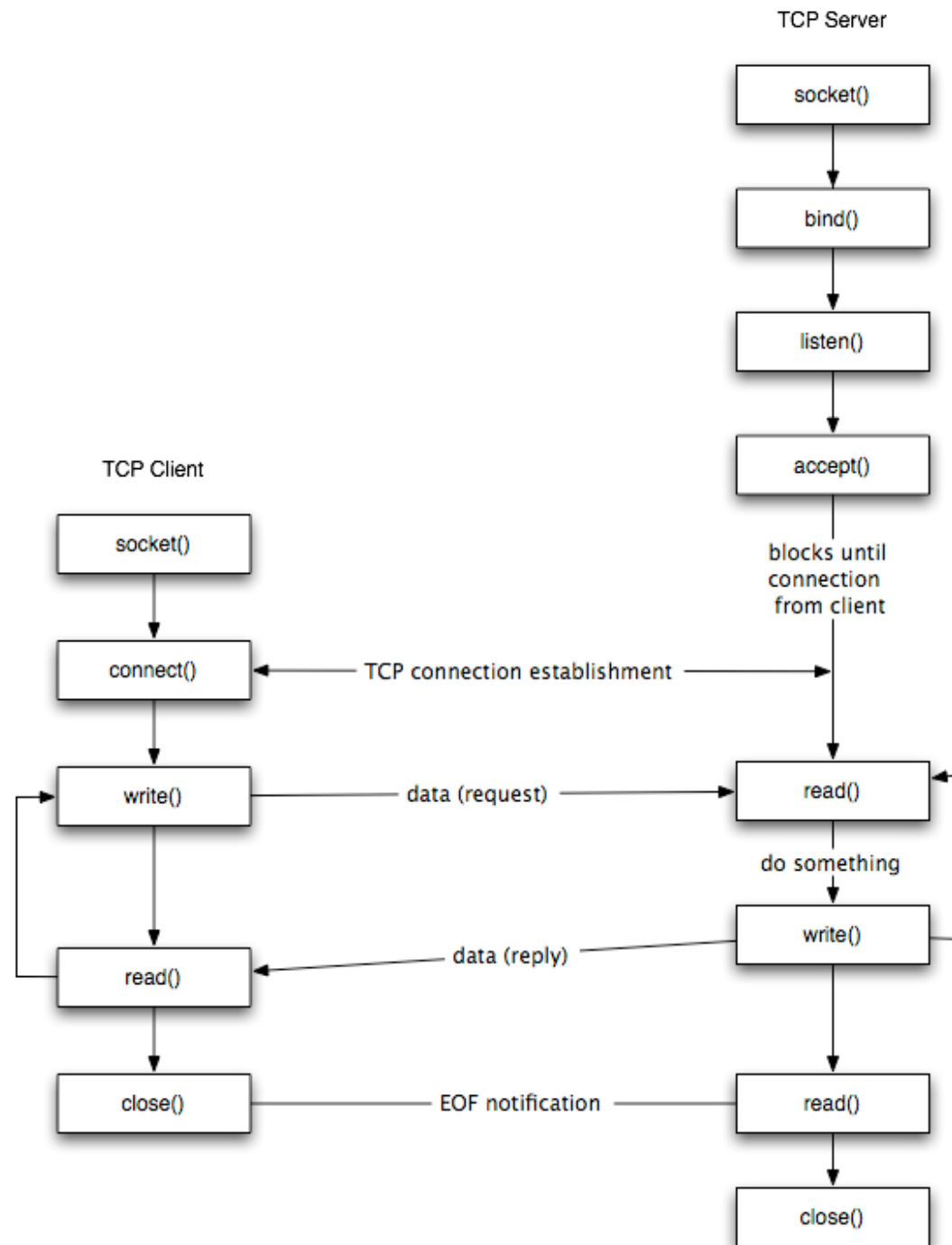
Ports

- Integer numbers usually between 1024 and 65535.
- All port numbers smaller than 1024 are reserved -- for example, http uses 80, ftp uses 21, and so on.
- Normally it is a practice to assign any port number more than 5000

Using Ports to Identify Services



TCP Socket Programming – Server and Client



Note:

send() can be used instead of write()
recv() can be used instead of read()

Check code examples attached with lab